

Exploratory Data Analysis

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Contents

Exploring the Data Answer the following questions:

What is your outcome variable(s)? How well does it measure the outcome you are interested? How does it relate to your expectations? What are your key explanatory variables? In addition, create a table of summary statistics for the variables you are planning to use.

Data Wrangling and Transformation:

What data cleaning did you have to do? How did you wrangle the data? Are you deciding to exclude any observations? If so, why? Did you have to create any new variables from existing variables? If so, how and why?

Codebook:

You must add a codebook – a description of all variables you are using, including ones you are creating for this project – to the README.md page of the data/ folder of your repo.

Data Visualization:

You must include at least 4 visualizations of your data. You must include your outcome variable in at least two plots and your key explanatory variable in at least two of these plots. You must use visualizations that are appropriate for the data type (categorical vs numeric, continuous vs discrete) of your outcome and explanatory variables. For example, you should not use a histogram to plot a categorical variable.

(OPTIONAL) Data Analysis:

If you would like, you can start to sketch out some data analysis/modeling. This will not be counted for or against you, but allows you to get ahead and for me and your peers to provide early feedback.

Formatting:

All parts of the Exploratory Data Analysis should be professionally formatted. For example, this means labeling plots and figures. The final product should be a pdf or html file rendered from a quarto file. You must comment your code! You must suppress all warnings and messages. All plots must be professional in appearance, including meaningful axes and legend label and titles. For this Exploratory Analysis assignment, please display your code in the rendered output. Style and format do count for this assignment, so please take the time to make sure that everything looks good and that your data and code are properly formatted.